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Graded Assignment • 30 min

Due Mar 23, 11:59 PM CET

1. Which of the following statements is not correct about LTE OFDM? 1 point

- OFDM is a parallel transmission method using multiple subcarriers.
- 1200 subcarriers with 15KHz subcarrier spacing are used when the system bandwidth is 20MHz.
- The OFDM symbol length, which is the inverse of the system bandwidth, is 50ns when the system bandwidth is 20MHz.
- The cyclic prefix helps to keep the orthogonality among the multiple subcarriers in most typical practical cellular environments.

2. Which of the following statements is not correct about LTE MCS? 1 point

- BPSK, QPSK, 16QAM, and 64QAM can be used for each subcarrier in an OFDM symbol.
- Turbo code with various code rates is used as the channel code in LTE.
- Each LTE MCS is mapped to its own channel quality information index.
- The data rate of an LTE MCS is determined by the modulation order.

3. Select the most appropriate one for the procedure of LTE H-ARQ. 1 point

- (CQI report) → (Scheduling) → (Data transmission) → (ACK or NACK) → (Retransmission if NACK)
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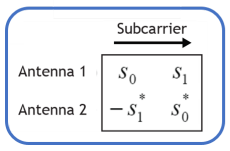
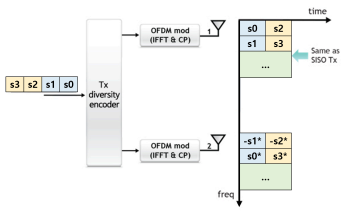
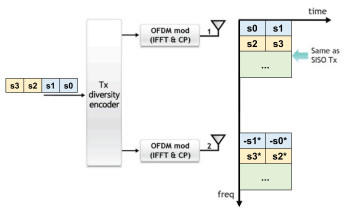
4. Which of following statements is not correct about LTE frame structure and multiple access? 1 point

- The basic resource unit is 1ms in time and 12 subcarriers in frequency.
- In downlink, orthogonal frequency division multiple access is used.
- In uplink, every user transmits with the same transmission power.
- In uplink, single carrier frequency division multiple access is used.

5. Which of the Tx structure is appropriate for the following scheme of SFBC (Space-Frequency-Block-Coding) ? 1 point



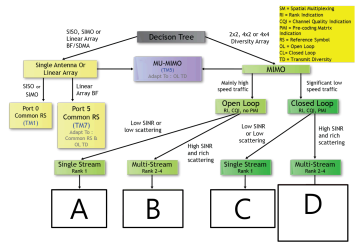
- 

- 

6. Which of the following statements does not correspond to LTE-Advanced key items? 1 point

- Coordinated multipoint transmission/reception, called CoMP.
- Peak spectrum efficiency of 20 bps / Hz for downlink, and 30 bps / Hz for uplink.
- Extension of downlink diversity to 8 antennas.
- Uplink diversity utilizing up to 4 antennas.
- Further enhancement of downlink MU-MIMO (R10 main issue).

7. Which of the following is incorrect? 1 point



- A: Transmit diversity (TM2)
- B: Closed loop spatial multiplexing (TM3)
- C: CL rank = 1 (TM6)
- D: Open loop spatial multiplexing (TM4)

8. What is the main reason to have open loop power control? 1 point

- Fast set-up for initial power level.
- Fast support of moving users.

9. Why is PF scheduler "proportionally" fair? 1 point

- It prioritizes less received users.
- It always gives more chances to users closer to the base station.